



Strain mapping in the Achilles tendon – A systematic review



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ABSTRACT

Achilles tendinopathy remains one of the most prevalent overuse injuries in elite as well as recreational athletes. Regardless of the fact that the aetiology of tendinopathy has not been fully understood, therapeutic mechanical loading programs have emerged as being the treatment of choice. In this light, mechanical properties of the tendon and their response to changes in loading or unloading have been the subject of many previous investigations. One of these properties often investigated is strain, a measure of relative deformation. By means of a systematic review, an overview was given of research in this field, with a primary objective to list the methods used and secondary aim to synthesize data on strain mapping in the Achilles tendon. Following the guidelines of the PRISMA statement, 47 articles were found appropriate for qualitative assessment. Achilles tendon strain has been investigated across a variety of contexts, including the response to exercise, walking, unloading, ageing, hormonal changes and weight. Only three studies investigated the effect of the presence of tendinopathy on strain. Ultrasound was the most often used imaging modality to measure or estimate strain. Further methodological parameters, e.g. the location of measurement, differed greatly between all different studies. Nearly all studies considered global strain. Some studies investigated the transverse strain response of the Achilles tendon. Recently, however, the role of local – intratendinous – strain distribution has been found to be of critical importance and further studies should focus on imaging modalities to investigate these local changes.

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1. Introduction

Tendinopathy is known to be the Achilles heel for both athletes and clinicians. Although the Achilles tendon is the thickest and strongest tendon in the human body, its structure is often affected. Achilles tendinopathy – the clinical syndrome characterized by a combination of pain and swelling in and around the Achilles tendon, accompanied by impaired performance (Khan et al., 2002; Maffulli et al., 1998) – is the most common overuse injury of the Achilles tendon, representing 55–65% of the injuries in this tendon (Järvinen et al., 2005). Furthermore, the incidence is still increasing as a result of the rising recreational and competitive sports participation in the population.

Many risk factors, both extrinsic and intrinsic, may predispose to injury, but pathoetiology remains poorly understood. Tendons play an important role as a force transducer from the muscle to the bone and are designed to withstand considerable loads. Mechanical loading of tendon tissue is however an important cause as well

as potential therapeutic goal in case of tendinopathy (Magnusson et al., 2010; Khan and Scott, 2009). Strain induced by this mechanical loading seems to be one of the key aspects (Maganaris et al., 2008).

Strain is a description of mechanical deformation, referring to the relative differences in particle displacement within a material. In tendon mechanics, the muscle contraction is an external force that creates a local stress field between the muscle and the bone attachment, which then induces strain (Józsa and Kannus, 1997). The relation between stress and strain is well described in mechanics using a so-called elastic modulus. In the case of linear elastic materials, this elastic modulus is known as Young's modulus, describing a linear relationship between stress and strain. Young's modulus can thus be regarded as a measure for the 'stiffness' of a material. The mechanical properties of tissues can be described using either material characteristics (e.g. Young's modulus, if we assume it to be linear elastic) or using a description of the deformation under stress itself (e.g. strain) (Beer et al., 2009). As opposed to Young's modulus, strain can be measured non-invasively and is thus the preferred means to characterize the biomechanical properties of tissue (Seynnes et al., 2015).

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The measurement or estimation of strain has been the target of many previous studies during the last decades. The methodologies of these different studies have however been heterogeneous in for instance the imaging technique used (Magnusson et al., 2001; Finni et al., 2003), the population studied (Wearing et al., 2013; Farris et al., 2012) or the effect investigated (Stafilidis and Arampatzis, 2007; Lee et al., 2006). This systematic review will give an overview of the research in this field, with a primary objective to list the methods used and secondary aim to synthesize data on strain mapping in the Achilles tendon.

2. Methods

This systematic review was conducted following the guidelines of the PRISMA statement (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) (Moher et al., 2009).

2.1. Search

Relevant studies were identified conducting a computer-based literature search in the MEDLINE database using PubMed. The following main keywords were used: “Achilles tendon”, “Tendinopathy” and “Strain”. These keywords were then expanded to aim for an inclusion of all the relevant literature (see Fig. 1).

2.2. Study selection

Only studies published in English were considered. In addition, ‘humans’ was marked as a limit for ‘species’. Studies involving subjects with normal Achilles tendons as well as subjects with Achilles tendinosis were included. Only in vivo studies were included. Studies involving subjects with an Achilles tendon rupture and/or subjects who underwent Achilles tendon surgery were not covered by this systematic literature search. Studies concerning subjects with other diseases or disorders such as neurological, systemic or neuromuscular problems were beyond the scope of this research work.

2.3. Review process

In the first stage of selection, all titles and abstracts were assessed for inclusion using the selection criteria. The primary reviewers (S.B. and H.D.) assessed the content of all full text articles, making the final inclusion/exclusion decision by consensus of three reviewers (S.B., H.D. and K.P.). There was no blinding to study author, place of publication or results.

2.4. Data items

Key data were extracted from each study. This included measurement technique, location, procedure and strain direction; population; and Achilles tendon strain values (longitudinal or transverse).

3. Results

The literature search was conducted on the 17th of October 2014 and updated the 21st of August 2015. It yielded 1901 articles to be screened. Sixty-four articles were eligible for full text assessment. Of these 64 articles, 47 articles were found to be appropriate for qualitative assessment. The full PRISMA flow diagram is depicted in Fig. 2.

4. Technical aspects

4.1. MRI versus ultrasound

Real-time 2D-ultrasound imaging is the most often used medical imaging technique to assess Achilles tendon strain. Only four of the included studies used MRI to visualize the reference points of interest in order to quantify tissue lengthening and shortening (Finni et al., 2003; Lee et al., 2006; Iwanuma

1) Achilles tendon: achilles tendon OR calcaneal tendon OR triceps surae

2) Tendinopathy: tendinopathy OR tendinosis OR tendinitis

3) Strain: strain OR stress OR tension OR elasticity OR Young's modulus OR elongation OR lengthening

To obtain the desired content for this literature search, the above-mentioned expansions were combined as follows:

[(1) AND (3)] OR [(2) AND (3)]. Thus, the complete search consisted of the following query:

[(Achilles tendon OR calcaneal tendon OR triceps surae) AND (strain OR stress OR tension OR elasticity OR Young's modulus OR elongation OR lengthening)] OR [(tendinopathy OR tendinosis OR tendinitis) AND (strain OR stress OR tension OR elasticity OR Young's modulus OR elongation OR lengthening)]

Fig. 1. Search terms.

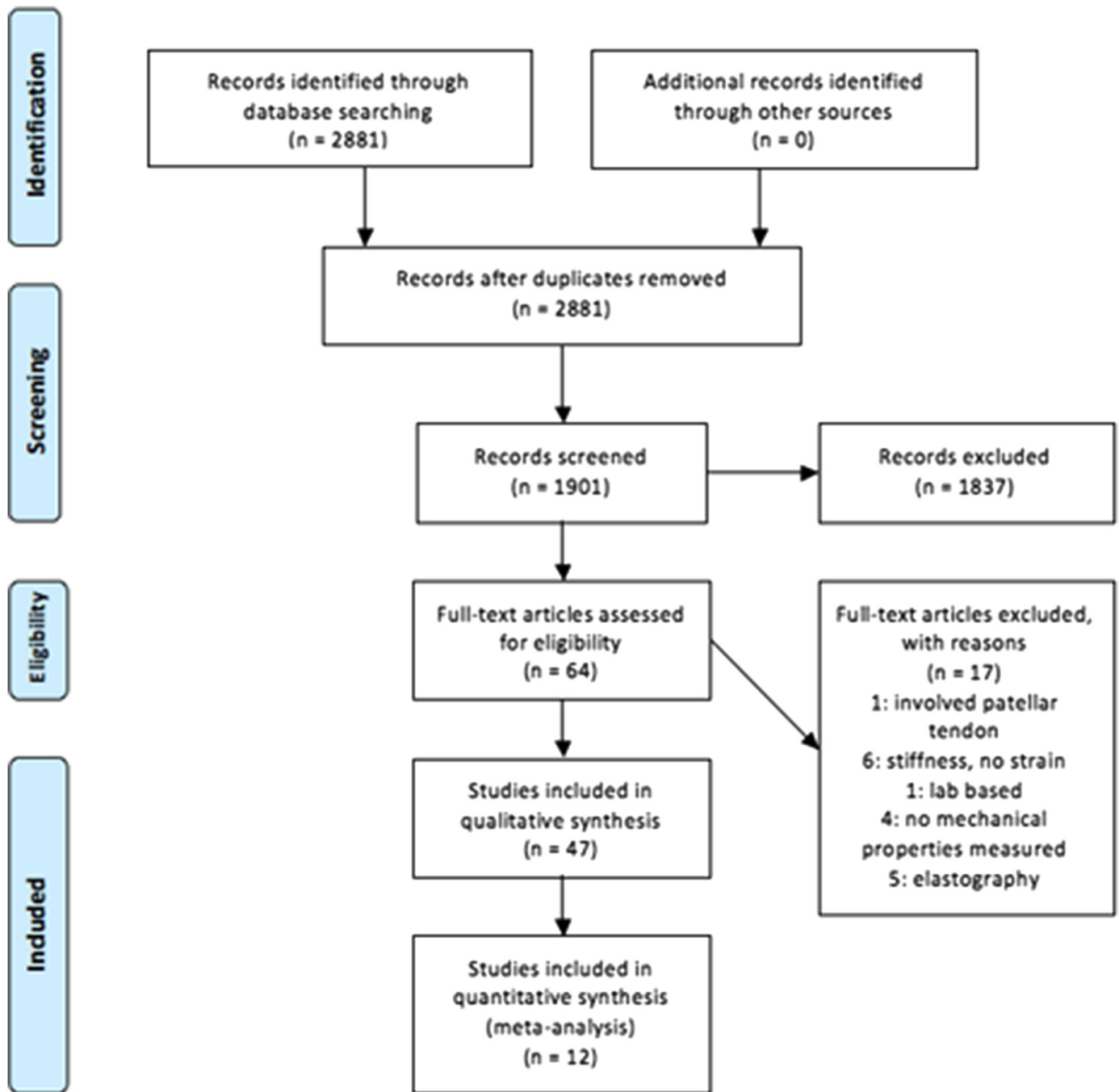


Fig. 2. PRISMA flow diagram.

Table 1
MRI based studies.

Study	Location	Procedure	Population
Iwanuma (2011)	Aponeurosis and free Achilles tendon	Active contraction (30 & 60% MVC)	6 men and 6 women
Shin (2008)	Whole tendon	Active contraction (40% MVC)	5 men
Lee (2006)	Soleus MTJ – calcaneal insertion	Active contraction (20% MVC)	3 men and 5 women
Finni (2003)	Soleus MTJ – calcaneal insertion	Active contraction (20 & 40% MVC)	4 men and 4 women

Table 2
3D-ultrasound based studies.

Study	Location	Procedure	Population
Farris (2013)	Distal and proximal Achilles tendon	Active contraction (50% MVC)	6 men and 5 women
Lichtwark (2013)	Myotendinous junction and free Achilles tendon	Active contraction (50% MVC)	9 men
Obst et al. (2013)	Whole tendon	Active contraction (70% MVC)	8 men

et al., 2011; Shin et al., 2008). Three studies used 3D ultrasound as imaging technique to determine longitudinal and transverse strain (Farris et al., 2013; Obst et al., 2014; Lichtwark et al., 2013). Details of the above mentioned studies using MRI and 3D-ultrasound are listed in Tables 1 and 2.

4.2. Longitudinal versus transverse

Strain direction was measured longitudinally in the vast majority of cases. However, in seven studies, strain was also measured in a transverse direction (Grigg et al., 2012; Wearing et al., 2011; Grigg et al., 2010; Farris et al., 2013; Iwanuma et al., 2011; Obst et al., 2014; Wearing et al., 2013). It is noteworthy that strain values were different depending on the strain direction; positive strains were recorded in the longitudinal direction and negative strains in the transverse direction. In addition, all included studies dealt with global Achilles tendon strain measurements, mostly using the gastrocnemius-Achilles musculotendinous junction and calcaneal insertion as reference points for strain tracking. None of the studies measured the local strain differences within the tendinous structure of the Achilles tendon.

4.3. Procedure

The preferred activation procedure to obtain a quantitative measure of Achilles tendon strain is maximum voluntary isometric contraction (MVIC). Further, three studies (Park et al., 2011; De Monte et al., 2006; Kawakami et al., 2008) used passive elongation as testing procedure and two studies (Grigg et al., 2012, 2010) examined the strain values during rest in a neutral ankle position (0° of dorsiflexion). These last authors investigate a “strain response”, being the change in tendon diameter over the course of a day, thus after loading. This is different from the strain investigated by other authors, being the relative change in length of the tendon during a contraction or passive elongation. Also, the amount of active contraction or passive elongation has to be considered when interpreting strain values. Fig. 3 shows the relation between the percentage of maximum voluntary contraction and measured strain values in three different studies (Iwanuma et al., 2011; Finni et al., 2003; Arampatzis et al., 2007a). This shows that a higher level of contraction leads to higher level of strain in the Achilles tendon. A further analysis and comparison between the results of these 3 studies is not possible due to the differences in methodology.

4.4. Location of measurement

Nomenclature is important, but is in this anatomical area sometimes unclear. For example, Achilles tendon, aponeurosis, free tendon, gastrocnemius tendon, myotendinous junction and

many other, are sometimes used in conflicting manner between different studies, which may cause misunderstandings and difficulty to compare strain values. There are differences in the strain values along the length of the tendon, in the different anatomical regions, for instance before and after the soleus joins the gastrocnemius tendon. Seven studies have investigated the comparison between strain in the aponeurosis versus strain in the Achilles tendon. This has shown that strain in the Achilles tendon is higher than in the aponeurosis region (see Fig. 4).

5. Physiological aspects

5.1. Effect of loading

Thirteen studies considered the effects of an applied load on the strain behavior of the Achilles tendon through exercise (Arampatzis et al., 2010, 2007a; Finni et al., 2003; Grigg et al., 2012; Hawkins et al., 2009; Lichtwark and Wilson, 2005; Mademli and Arampatzis, 2008; Peltonen et al., 2010; Orlando and Hawkins, 2007; Wearing et al., 2011; Iwanuma et al., 2011; Jeong et al., 2014; Magnusson et al., 2001). In addition to the effects of exercise on Achilles tendon strain, several studies examined Achilles tendon strain in relation to walking or sports activities (Arampatzis et al., 2007a, 2007b; Karamanidis and Arampatzis, 2006; Lichtwark and Wilson, 2006; Park et al., 2011; Rosager et al., 2002; Staflidis and Arampatzis, 2007; Farris et al., 2012; Lichtwark et al., 2013). Furthermore, the effects of chronic unloading on tendon strain distribution were examined as well (Lee et al., 2006; Shin et al., 2008).

After further selection of 8 studies investigating the effect of loading in a longitudinal design, a meta-analysis was performed. Four of these studies investigated two different subgroups, making the total 12 included groups (see Fig. 5). No statistically significant effect was found of loading on Achilles tendon strain values pre-versus post-intervention.

5.2. Influencing factors

Achilles tendon strain behavior has been investigated across a variety of contexts. Some studies (Hawkins et al., 2009; Mademli and Arampatzis, 2008; Karamanidis and Arampatzis, 2006; Kubo et al., 2007; Waugh et al., 2012; Neugebauer and Hawkins, 2012; Stenroth et al., 2012) examined the age-related differences in mechanical properties of the Achilles tendon, the effect of oestrogen on the strain behavior of the Achilles tendon (Bryant et al., 2008) and the effect of weight (Wearing et al., 2013).

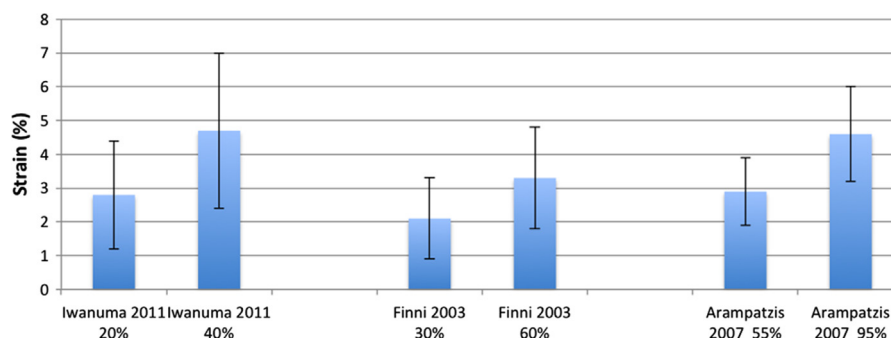


Fig. 3. The relation between % MVC and strain in three different studies.

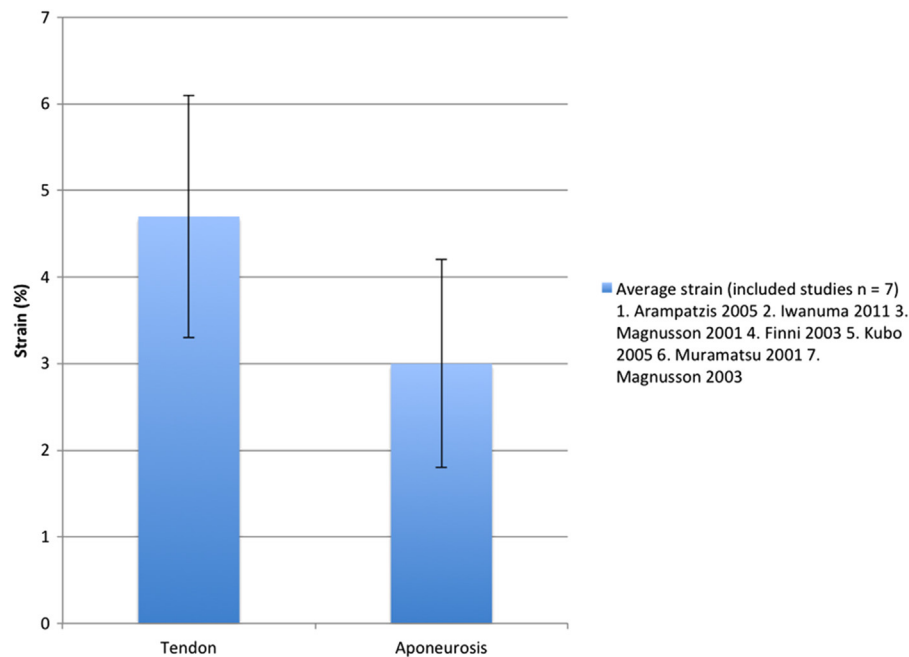


Fig. 4. Tendon versus aponeurosis strain (Arampatzis et al., 2005; Iwanuma et al., 2011; Magnusson et al., 2001; Finni et al., 2003; Kubo et al., 2005; Muramatsu, 2001; Magnusson et al., 2003).

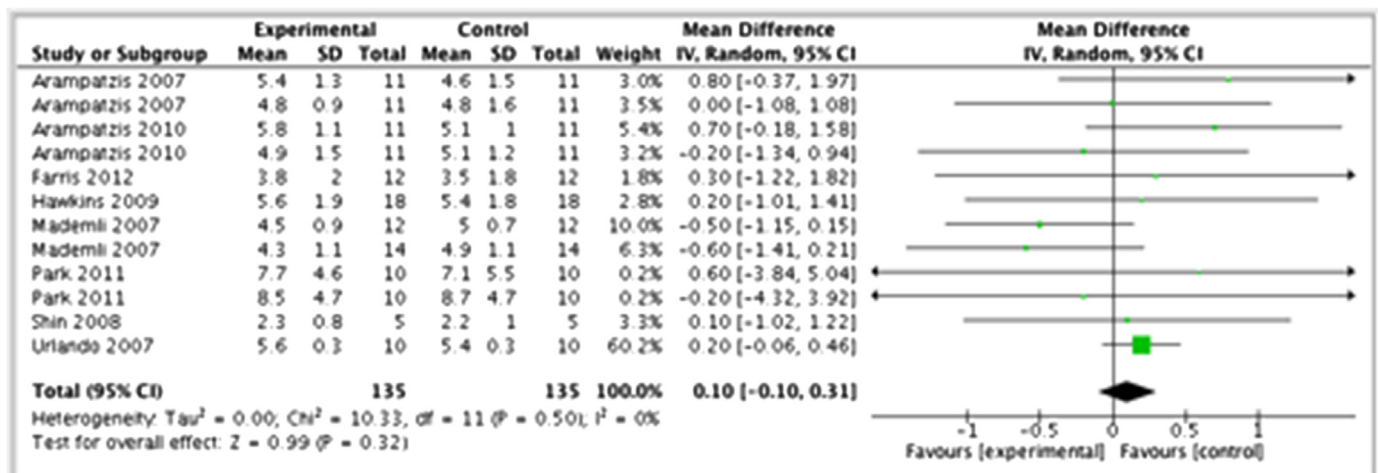


Fig. 5. Forest plot – Effect of loading on strain – The experimental group are the strain values after loading/exercise. The control group are the strain values before loading/exercise.

5.3. Tendinopathy

Only three research groups considered the in vivo Achilles tendon strain in the presence of Achilles tendinopathy (Grigg et al., 2012; Arya and Kulig, 2010; Child et al., 2010). Arya and Kulig found a 15% higher strain ($P \leq 0.05$) in the Achilles tendinopathy group ($5.14 \pm 0.57\%$) compared with the control group ($4.36 \pm 0.31\%$). The outcome of the study of Child et al. was in line with the previous study: the Achilles tendon strain was significantly higher ($P \leq 0.05$) in male athletes with Achilles tendinopathy ($5.2 \pm 2.6\%$) compared with those without it ($3.4 \pm 1.8\%$). In both studies, strain was measured in a longitudinal direction and the Achilles tendon length was defined as the distance between the insertion on the calcaneus and the medial gastrocnemius myotendinous junction. On the other hand, Grigg et al. determined the anteroposterior (transverse) strain 40 mm proximal to the calcaneal insertion in response to eccentric exercise. Eccentric exercises resulted in anteroposterior (transverse) strains

of -14.6% , -13.0% and -8.3% for the control, asymptomatic and symptomatic tendons, respectively. Participants in the symptomatic tendinopathy group were found to have a significantly ($P < 0.05$) lower strain when compared with pain-free counterparts and controls.

6. Discussion

6.1. Key findings

This systematic review gives an overview of research in the field of strain mapping in the Achilles tendon, with a primary objective to list the methods used and synthesize data on strain in the Achilles tendon. The technique most often used is real-time 2D ultrasound with tracking of two points (e.g. musculotendinous junction and calcaneal insertion) during an active maximal voluntary contraction. The relative change in tendon length then

leads to an estimation of tendon strain (expressed as a percentage). Average strain values range from 1.1% (Iwanuma et al., 2011) to 9.2% (Vaughn et al., 2012). Maximal absolute transverse strain was $-19.5 \pm 7.4\%$ (Wearing et al., 2011). Given the heterogeneous aspect of research in this field (with regard to population, procedure of activation, location of measurement, effect investigated) only the effect of loading was investigated by means of a meta-analysis on 12 subgroups (see Fig. 5). This showed no statistically significant effect on strain values in the Achilles tendon.

6.2. Technical aspects

6.2.1. Ultrasound versus MRI

Concerning the applied measurement technique of the included studies, initial observations showed real-time ultrasound scanning (two-dimensional) has been used in the majority of the cases as the preferred non-invasive measurement technique in the assessment of mechanical properties of the Achilles tendon complex. This technique allows in vivo tendon strain measurement mostly by tracking a reference point within the field of view. Multiple arguments support the choice for using US as the preferred measurement technique to assess mechanical tendon properties. Firstly, unlike MRI, an ultrasound device is readily available and overall less costly. Furthermore, the dynamic aspect during measurements is still more advanced when compared to MRI, although recent developments in cardiac imaging with 4D (3D+time) MRI look promising (Stankovic et al., 2014). Lastly, spatial resolution is better in ultrasound images, which is needed when aiming to use speckle tracking to measure strain (see further), which tends to use the secondary or tertiary fiber bundles of the tendon as reference point (Arndt et al., 2012). In their excellent paper, Seynnes et al. (2015) highlight the most important factors to take into consideration when investigating mechanical properties of tendons by means of ultrasound.

6.2.2. Other techniques

On-going research in the field of strain mapping in tendons has led to some new and interesting findings. Firstly, fiber optic sensors appear to be a promising new tool (Roriz et al., 2014). These reflect a narrow bandwidth of light, which responds faithfully to strain and temperature. The downside to this technique remains its minimally invasive nature, making it hard to be used in vivo and clinical practice. This technique can however be useful in research setting as a ground-truth for ex vivo studies trying to validate new non-invasive techniques. Secondly, computational studies have examined global and more recently local strain in Achilles tendon. These studies use biomechanical and mathematical models to calculate local strain (Ahmadzadeh et al., 2015). The use of these models has become an important research area in the field of biomechanics, but validation of individual models remains an important issue to take into consideration (Uchida et al., 2014). Since these models are often compared with or validated against ex vivo material or isolated tendons (Shearer, 2015), these studies did not fall within the scope of this review.

6.2.3. Procedure

Given the biomechanical characteristics of tendinous tissue, a distinction was made between testing procedures using active contraction and those using passive elongation as activation procedure to obtain Achilles tendon strain. Thus far, Achilles tendon strain has been investigated through active contraction in the vast majority of the included studies. However, the load applied on the Achilles tendon varied greatly between the different studies. Since many injuries occur during sports activities, it may be desirable to impose loads mimicking Achilles tendon strain during sports, which tend to be at higher speeds of movement and higher

absolute load (Stafilidis and Arampatzis, 2007). Also, whether or not there was preconditioning of the Achilles tendon may influence the mechanical properties measured, given the fact that a tendon behaves as a viscoelastic structure (Maganaris et al., 2008; Maganaris and Paul, 2002). Lastly, another important factor is ankle joint rotation. During experiments with voluntary contraction, when the foot moves into plantar flexion, there is always some degree of ankle joint rotation, which influences elongation of muscle and tendon (Maganaris, 2005). Since this change in position of the reference points for tracking is not taken fully into account, this can lead to an overestimation of strain. Farris et al. (2013) solved this by making sure the dorsal aspect of the foot always stayed in contact with the foot plate and have taken correction factors into account for passive displacement. Even then, Arampatzis et al. (2006) have shown that an absolute difference in calculated strain may be as high as 0.3%.

6.2.4. Location of measurement and role of local strain

All included studies in this review concerned global strain in which the strain has usually been measured in a longitudinal direction and the musculotendinous and osteotendinous junction were selected as reference points for strain tracking. Global strain has provided us with plenty of information regarding the behavior of tendon tissue in vivo, but there is however a need for further investigation on local – intratendinous – level. Sub-regional strain differences appear on cadaveric Achilles tendons (Lersch et al., 2012; Wren et al., 2001; Lyman, 2004). Lyman et al. found that strain in the posterior sites of the distal Achilles tendon increased significantly as the movement into dorsiflexion occurred, whereas the anterior sites showed a trend towards a decrease in strain. In addition, Lersch et al. found intratendinous strain differences on cadaveric Achilles tendons during different muscle loading conditions and at different calcaneus positions. Finally, Wren et al. investigated strain behavior of cadaveric Achilles tendons during repetitive cyclic loading to failure. They found different strain distributions within the proximal and distal regions of the Achilles tendons. Based on the findings of these studies on cadaveric Achilles tendons, it may be concluded that knowledge on local – intratendinous strain differences in the Achilles tendon in vivo is valuable.

Further supporting these needs for local strain research are some laboratory-based studies. Arnoczky et al. (2002) microscopically investigated rat tail cells with staining of the cell nucleus and witnessed an in situ cell nucleus deformation when under tensile load. Cheng and Screen (2007) looked at bleached collagen fibers and fibrils and saw an inhomogeneous strain response in the matrix. They found a large variation between samples, but overall found the local strain values to be much smaller than the applied values of strain in the loading axis, which would be suggestive for compression, fiber sliding and rotation of the collagen. Furthermore, in two different equine based studies (Thorpe et al., 2013a, 2013b), evidence was found for a local and rotational component in tendon movement, hypothesized by the authors as providing a better recoil and that way a possibly enhanced energy saving system. Lastly, Arndt et al. (2012) investigated healthy tendons during passive dorsiflexion and found a difference in strain between deep, middle and superficial parts of the Achilles tendon.

6.3. Physiological aspects

The meta-analysis performed in this paper (see Fig. 5) shows no effect of loading programs or exercise on the strain values measured. An important fact to take into consideration however is the heterogeneity in the definition of “loading”. Table 3 gives an overview of the differences in duration, volume and intensity of

the loading parameters in the included studies. The methodological design of these studies differs, but since the effect is measured using the same methodology this does not influence the comparison between the effect measured in the studies. It should however be noted again, that the methodology of strain measurement has its limitations and that small changes in strain will be difficult to detect.

6.3.1. Mechanical loading

The effect of loading on mechanical parameters of tendinous tissue has been the subject of many previous studies. Two interesting studies by the group of Arampatzis et al. have investigated the response of tendon to cyclic loading at different magnitudes and frequencies of the loading stimuli. Firstly, there appears to be a threshold of strain magnitude of mechanical loading, which has to be exceeded in order to influence the homeostasis of the tendon (Arampatzis et al., 2007a). This was shown by the fact that after a 14-weeks exercise intervention, only in the leg exercised at high strain magnitude changes were found in the mechanical parameters. Secondly, surprisingly, a threefold increase in the strain frequency (from 0.17 to 0.5 Hz) caused lower adaptational effects on the Achilles tendon mechanical and morphological properties (Arampatzis et al., 2010). Given the fact that the mechanical load can only cause changes in these properties by triggering cells and matrix or affecting collagen, the authors hypothesize that a lower strain frequency may lead to more effective fiber recruitment and therefore a greater transfer of the external tendon strain magnitude to the cellular level. In another longitudinal study, Orlando and Hawkins (2007) investigated the effect of an 8-week strength program. This showed no difference in strain of the Achilles tendon, despite a significant change in muscle strength. The authors hypothesize that tendons have a preferred strain limit that is maintained despite changes in triceps surae strength. It remains to

be seen for which reasons and through what exact mechanisms a tendon reacts to loading after an acute bout of exercise as well as in more chronic (over)loading conditions. Changes could be a physiological reaction to reduce stress as well as an attempt to repair damaged tissue (Magnusson et al., 2010) and might be a structural as well as mechanical adaptation. As mentioned earlier, also ageing seems to have an influence on tendinous material with overall lower strain levels measured in the elderly (Fig. 6). Although no statistically significant finding, this conflicts with the overall perception that stiffness decreases during ageing due to worsening material properties, which would contrastingly lead to higher strain values. However, as mentioned earlier, it is important to take into consideration that great inter-individual differences exist, leading to difficulties to interpret results of cross-sectional studies. Stenroth et al. (2012) suggest that regardless of age, Achilles tendon mechanical properties adapt to match the level of muscle performance. Unfortunately, no studies have investigated the change in strain values in response to ageing in a longitudinal design longer than 6 months. Given the fact that mechanical properties of tendon material appear mostly related to muscle strength with little influence of ageing or loading, it might be interesting to look at the tendon during growth. As was shown by Heinemeier et al. (2013), most of the tendon core is formed during height growth, reaching a steady state around the age of 17. This leads to the hypothesis that the influence of mechanical loading of tendinous tissue during growth could have a much more profound and possibly long-lasting effect on mechanical and structural properties, then any loading program or exercise during later stages in life.

6.4. Implications on further research

It is noteworthy that only three research groups considered the in vivo Achilles tendon strain in response to Achilles tendinopathy (Grigg et al., 2012; Arya and Kulig, 2010; Child et al., 2010). Since the ultimate objective is to help patients and athletes with Achilles tendinopathy through optimizing their therapeutic exercise program, future research regarding strain measurement on subjects with symptomatic and asymptomatic Achilles tendinosis is warranted.

Recently, the strain measurement technique using real-time ultrasound has been further automated using speckle-tracking algorithms. These allow tracking frame-to-frame displacements of speckle patterns within the US image. Strain rate imaging using two- and three-dimensional ultrasound speckle tracking has been used for several years in cardiology (D'Hooge et al., 2002) and

Table 3
Content of loading programs.

Shin (2008)	6 weeks rehab after 4 weeks unloading
Farris (2012)	30 min run
Peltonen (2010)	Hopping
Hawkins (2009)	7 min warm-up after 1 h immobilization
Orlando (2007)	8 weeks strength program
Mademli (2008)	40% MVC as long as possible on isokinetic device
Arampatzis (2010)	14 weeks cyclic load at a certain frequency
Arampatzis (2007)	14 weeks cyclic load at a certain magnitude
Park (2011)	6 min warm-up jog and 5 × 30 s static stretch

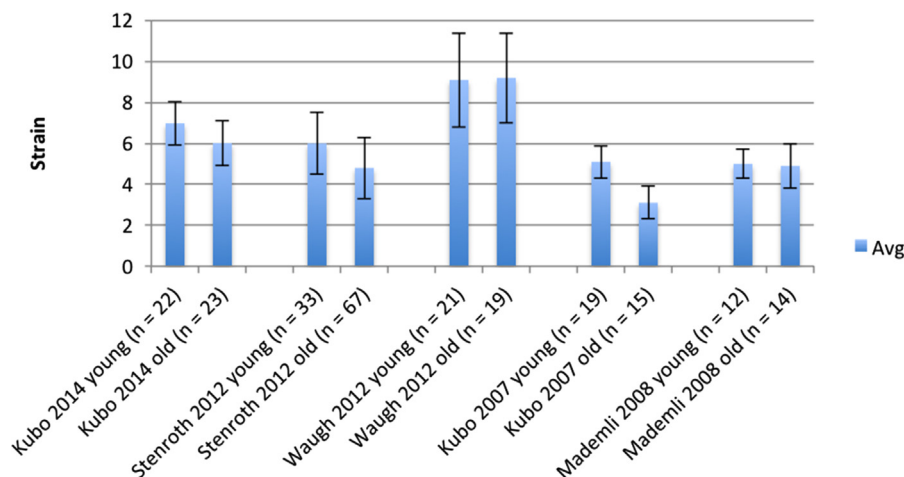


Fig. 6. Relation between age and strain (Kubo et al., 2014; Stenroth et al., 2012; Waugh et al., 2012; Kubo et al., 2007; Mademli and Arampatzis, 2008).

seems to be a promising technique in a number of musculoskeletal applications. Kim et al. (2011) described an in vivo strain analysis of the intact supraspinatus tendon by ultrasound speckle tracking imaging, but there are limitations to be taken into consideration (Slagmolen et al., 2012). Korstanje et al. (2010) evaluated the accuracy of tracking the flexor digitorum superficialis of a human cadaver hand and the study of Yoshii et al. (2011) demonstrated that an ultrasound method based on speckle tracking had the potential to evaluate finger flexor tendon motion in vivo. Although these studies have several limitations, they prove ultrasound speckle-tracking is a promising technique for in vivo strain analysis of tendons.

Conflict of interest statement

There are no conflicts of interest to declare that would be directly relevant to the content of this review.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.jbiomech.2016.02.057>.

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